

Multiple Choice (10 marks)

1. _____ is also a part of darknet that is employed for transferring files anonymously.
 - (a) Freenet
 - (b) ARPANET
 - (c) Stuxnet
 - (d) Internet
2. The _____ is categorized as an unknown segment of the Deep Web which has been purposely kept hidden & is inaccessible using standard web browsers.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Dark web
 - (d) Surface web
3. A/an _____ is a program that steals your logins & passwords for instant messaging applications.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) Trojan-Downloader
 - (d) Ransom Trojan
4. What does EXE do when the STP solver times out on a constraint query for a particular path?
 - (a) Assume that the query is satisfiable and continue executing the path.
 - (b) Assume that the query is not satisfiable and stop executing the path
 - (c) Restart STP and retry the query, up to a limited number of retries.
 - (d) Remove a subset of the constraints and retry the query.
5. _____ is the central node of 802.11 wireless operations.
 - (a) WPA
 - (b) Access Point
 - (c) WAP
 - (d) Access Port

True / False (10 marks)

Answer True or False

6. True or False: A session symmetric key can be reused multiple times for different communication sessions without compromising security.
7. True or False: The four primary security principles related to messages are Confidentiality, Integrity, Non-repudiation, and Authentication.
8. True or False: A ping sweep is used to identify live systems on a network by sending ICMP echo requests to multiple IP addresses.
9. True or False: A MAC that produces a fixed 5-bit tag can be insecure, as an attacker can guess the tag for messages due to limited possibilities.
10. True or False: WPA 2 is a method used to enhance wireless security by adding new devices without the need for a password.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the vulnerabilities and limitations of the WEP encryption standard in the context of cybersecurity, and analyze why it is considered the least secure option compared to other standards.
12. Discuss the concept of buffer-overflow vulnerabilities in application coding, including their potential impacts on system security and strategies for mitigation.

End of Paper